

Gaussian Approximation for Color Counts in a Graph with Multiple Sources

Problem setup

- Graph $G = (V, E)$ with a single sink node T (absorbing).
- K sources $S_1, \dots, S_K$.
- Each source S_k injects N_k particles/sec (large N_k), of *color* k .
- Particles do independent random walks and are absorbed at T .
- Consider node $A \neq T$.
- Let $X_k(A)$ be the number of color- k particles at A in steady state.

Goal: Approximate the *joint distribution* of

$$(X_1(A), X_2(A), \dots, X_K(A))$$

in steady state for large N_k .

Step 1: Mean number of particles

The expected number of color- k particles at A is

$$\lambda_k(A) = N_k \cdot H(S_k, A),$$

where $H(S_k, A)$ is the expected number of visits to A by a particle starting at S_k before hitting T .

Step 2: Distribution of color counts

Particles of color k move independently:

$$X_k(A) \sim \text{Poisson}(\lambda_k(A)).$$

Step 3: Joint distribution

Because colors are independent,

$$(X_1(A), \dots, X_K(A)) \sim \bigotimes_{k=1}^K \text{Poisson}(\lambda_k(A)).$$

Step 4: Large- λ_k Gaussian Approximation

For large $\lambda_k(A)$:

$$X_k(A) \approx \mathcal{N}(\lambda_k(A), \lambda_k(A)).$$

Thus jointly:

$$(X_1(A), \dots, X_K(A)) \approx \mathcal{N}(\mu(A), \Sigma(A)),$$

where

$$\mu(A) = \begin{bmatrix} \lambda_1(A) \\ \vdots \\ \lambda_K(A) \end{bmatrix}, \quad \Sigma(A) = \text{diag}(\lambda_1(A), \dots, \lambda_K(A)).$$

Step 5: Distribution over nodes

Suppose A is random with prior $P(A)$. Then:

$$(X_1, \dots, X_K) \mid A \sim \mathcal{N}(\mu(A), \Sigma(A)).$$

Joint distribution:

$$P(A, x) = P(A) \times \mathcal{N}(x \mid \mu(A), \Sigma(A)).$$

Marginal distribution over counts:

$$P(x) = \sum_A P(A) \mathcal{N}(x \mid \mu(A), \Sigma(A)).$$

Step 6: Mutual Information Approximation

The mutual information between the color-count vector and the node identity is:

$$I((X_1, \dots, X_K); A) = H(X_1, \dots, X_K) - \sum_A P(A) H(X_1, \dots, X_K \mid A).$$

Under the Gaussian approximation:

$$H(X_1, \dots, X_K \mid A) = \frac{K}{2} \log(2\pi e) + \frac{1}{2} \sum_{k=1}^K \log \lambda_k(A).$$

The marginal distribution is a *Gaussian mixture*:

$$P(x) = \sum_A P(A) \mathcal{N}(x \mid \mu(A), \Sigma(A)).$$

Its entropy $H(X_1, \dots, X_K)$ typically requires numerical estimation.

Step 7: Final Approximate Formula

For large N_k :

$$(X_1, \dots, X_K) \mid A \approx \mathcal{N}(\mu(A), \Sigma(A))$$

with

$$\mu_k(A) = N_k H(S_k, A), \quad \Sigma(A) = \text{diag}(\mu(A)).$$

Then:

$$I \approx H(\text{Gaussian mixture}) - \sum_A P(A) \frac{1}{2} \sum_{k=1}^K \log(2\pi e \lambda_k(A)).$$

Interpretation

- Mutual information quantifies how well the *mean color profile* $\mu(A)$ distinguishes nodes.
- If the vectors $\mu(A)$ are well separated across nodes (relative to their variances), mutual information is large.
- If $\mu(A)$ is nearly constant across nodes, mutual information is small.